Human escalation workflows

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete reviewer handoff and pending-state record.

Keep case, reported facts, source IDs, requested decision, owner, deadline, fallback and actual communication evidence together.

Keep one habit: Recall → W1–W3/W5 → independent deadline case → handoff project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Escalation routes unresolved decisions to an authorised person with sufficient context. It must define owner, deadline and fallback.

- Toy workflow routes conflicting room approval to coordinator with source IDs and requested decision. While awaiting review the assistant returns needs_review and performs no booking.

Incident investigation

> **My remembered idea, example and correction**

Release and rollback

> **My remembered idea, example and correction**

Caching and stale information

> **My remembered idea, example and correction**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Escalation routes unresolved decisions to an authorized person with sufficient minimized context, an owner, deadline and fallback. Main fictional room conflict: source S1 says B12, S2 says D4. With no reviewer, silently booking the first room is unauthorized.

  Assign the trusted coordinator, include case/source IDs and requested approval decision, and keep needs_review with no booking while awaiting a permitted decision. Added local deadline is virtual minute 30: equality is overdue but never approval. Wrong-role or unreachable owner leaves unresolved status. A handoff-ready packet is not evidence of sending, receipt, review or approval.

  Separate picture lesson-notes bot cannot certify a reported damaged bottle; preserve “learner reports damage, physical condition not verified” and use the designated supervisor route. If unreachable, the bot gains no authority; real institutions define actual responses. Only a separately authorized configured notification channel may be used.

  Code makes local fictional packets and decisions with empty effect/notification traces; it sends no messages, books nothing and provides no equipment verdict.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Escalation routes unresolved decisions to an authorized person with sufficient minimized context, an owner, deadline and fallback. Main fictional room conflict: source S1 says B12, S2 says D4. With no reviewer, silently booking the first room is unauthorized.

  Assign the trusted coordinator, include case/source IDs and requested approval decision, and keep needs_review with no booking while awaiting a permitted decision. Added local deadline is virtual minute 30: equality is overdue but never approval. Wrong-role or unreachable owner leaves unresolved status. A handoff-ready packet is not evidence of sending, receipt, review or approval.

  Separate picture lesson-notes bot cannot certify a reported damaged bottle; preserve "learner reports damage, physical condition not verified" and use the designated supervisor route. If unreachable, the bot gains no authority; real institutions define actual responses. Only a separately authorized configured notification channel may be used.

  Code makes local fictional packets and decisions with empty effect/notification traces; it sends no messages, books nothing and provides no equipment verdict.

- Conflict is detected, but no reviewer is assigned and the system silently books the first room. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Escalation routes unresolved decisions to an authorized person with sufficient minimized context, an owner, deadline and fallback. Main fictional room conflict: source S1 says B12, S2 says D4. With no reviewer, silently booking the first room is unauthorized.

  Assign the trusted coordinator, include case/source IDs and requested approval decision, and keep needs_review with no booking while awaiting a permitted decision. Added local deadline is virtual minute 30: equality is overdue but never approval. Wrong-role or unreachable owner leaves unresolved status. A handoff-ready packet is not evidence of sending, receipt, review or approval.

  Separate picture lesson-notes bot cannot certify a reported damaged bottle; preserve "learner reports damage, physical condition not verified" and use the designated supervisor route. If unreachable, the bot gains no authority; real institutions define actual responses. Only a separately authorized configured notification channel may be used.

  Code makes local fictional packets and decisions with empty effect/notification traces; it sends no messages, books nothing and provides no equipment verdict.

- A report omits this qualification: Escalation without a responsible actor and declared fallback cannot guarantee resolution. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Escalation routes unresolved decisions to an authorized person with sufficient minimized context, an owner, deadline and fallback. Main fictional room conflict: source S1 says B12, S2 says D4. With no reviewer, silently booking the first room is unauthorized.

  Assign the trusted coordinator, include case/source IDs and requested approval decision, and keep needs_review with no booking while awaiting a permitted decision. Added local deadline is virtual minute 30: equality is overdue but never approval. Wrong-role or unreachable owner leaves unresolved status. A handoff-ready packet is not evidence of sending, receipt, review or approval.

  Separate picture lesson-notes bot cannot certify a reported damaged bottle; preserve “learner reports damage, physical condition not verified” and use the designated supervisor route. If unreachable, the bot gains no authority; real institutions define actual responses. Only a separately authorized configured notification channel may be used.

  Code makes local fictional packets and decisions with empty effect/notification traces; it sends no messages, books nothing and provides no equipment verdict.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Reviewer deadline is 16:00; no reply by 16:00. Policy says remain pending, not auto-approve. Decide.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled human escalation workflows **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of human escalation workflows.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Human escalation workflows. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Human escalation workflows. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Reviewer deadline is 16:00; no reply by 16:00. Policy says remain pending, not auto-approve. Decide.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

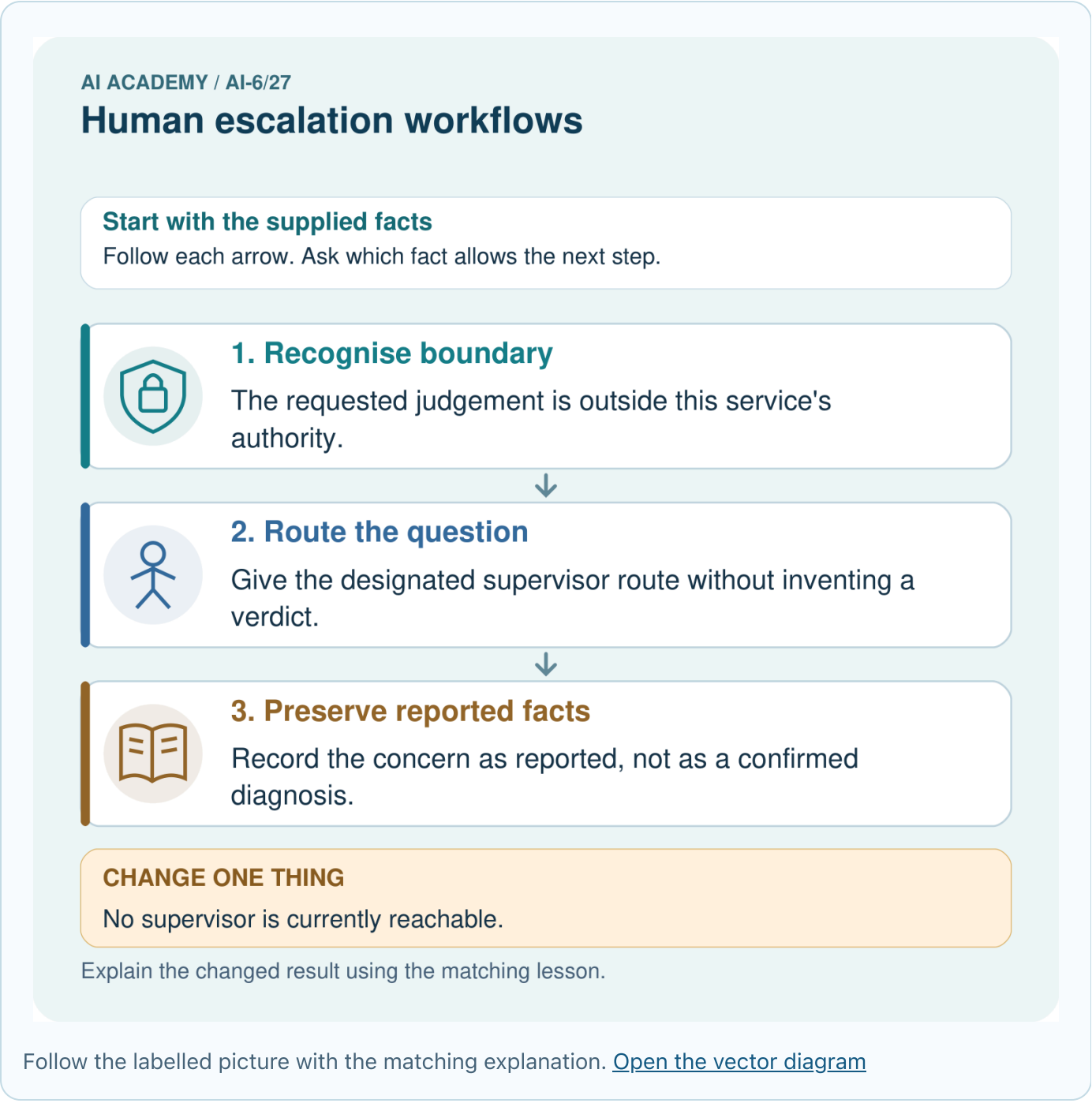
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A useful human handoff. A learner asks whether a damaged laboratory bottle is safe; the bot handles lesson notes only.
- The contract excludes equipment certification. An assigned adult supervisor handles equipment concerns.
- Identify the exact service boundary in the bottle question. What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A useful human handoff. A learner asks whether a damaged laboratory bottle is safe; the bot handles lesson notes only.
- Which labelled step matches this detail? Record the concern as reported, not as a confirmed diagnosis.
- 1. Recognise boundary
- 2. Route the question

- 3. Preserve reported facts

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Preserve reported facts Record the concern as reported, not as a confirmed diagnosis. Escalation needs an appropriate person and accurate context.

Model confidence cannot replace required authority or physical inspection.

Changed-case question and optional hint

- Changed-case practice: No supervisor is currently reachable. Which conclusion is supported by the supplied facts?

- 1. State unresolved status and use the institution’s approved waiting procedure.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: State unresolved status and use the institution’s approved waiting procedure. Lack of a human reply does not expand the bot’s authority.

Repair and retain the record

Review Incident investigation (ai-6/26). Goal: Explain the **mechanism** and terminology of incident investigation. Short practice: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?. Return to this week and explain what you changed.

Review Release and rollback (ai-6/24). Goal: Explain the **mechanism** and terminology of release and rollback. Short practice: Compute A’s and B’s correctness percentages.

Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?. Return to this week and explain what you changed.

Review Caching and stale information (ai-6/21). Goal: Explain the **mechanism** and terminology of caching and stale information. Short practice: Identify the version mismatch in the cached 09:00 answer.

What should the first version -5 request return, and what permits a later version -5 cache hit?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

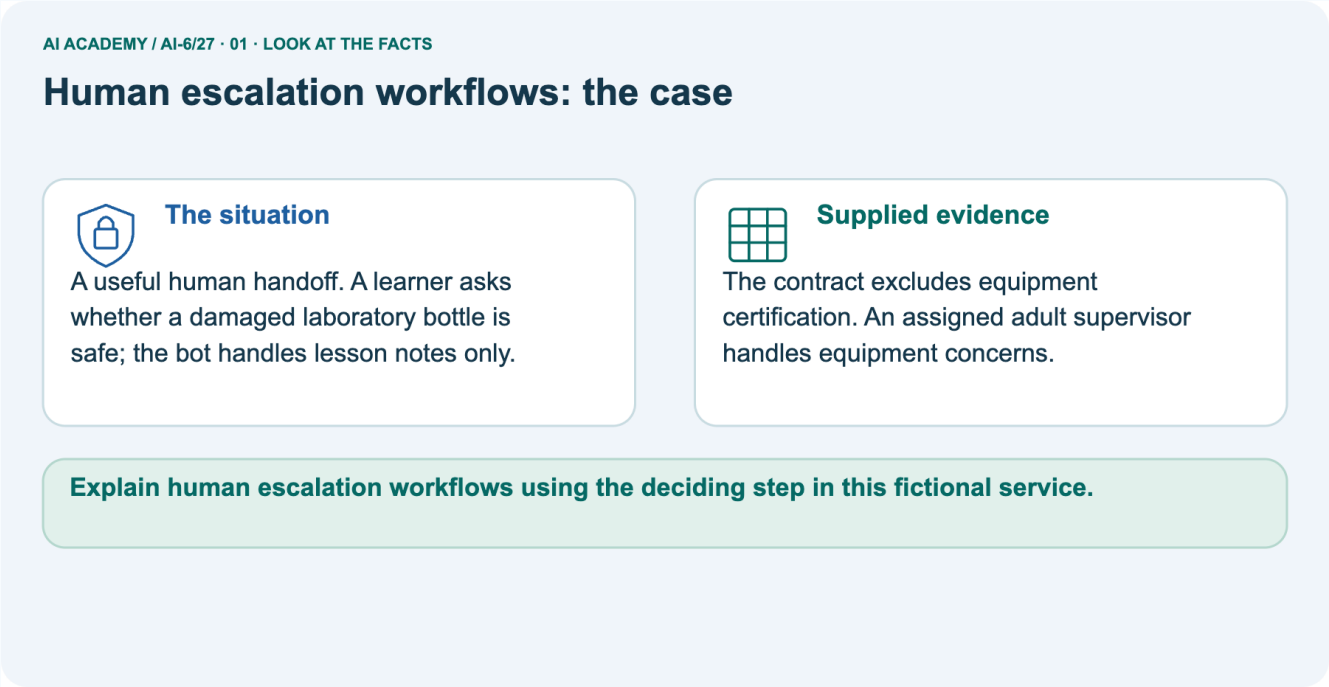
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Human escalation workflows: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

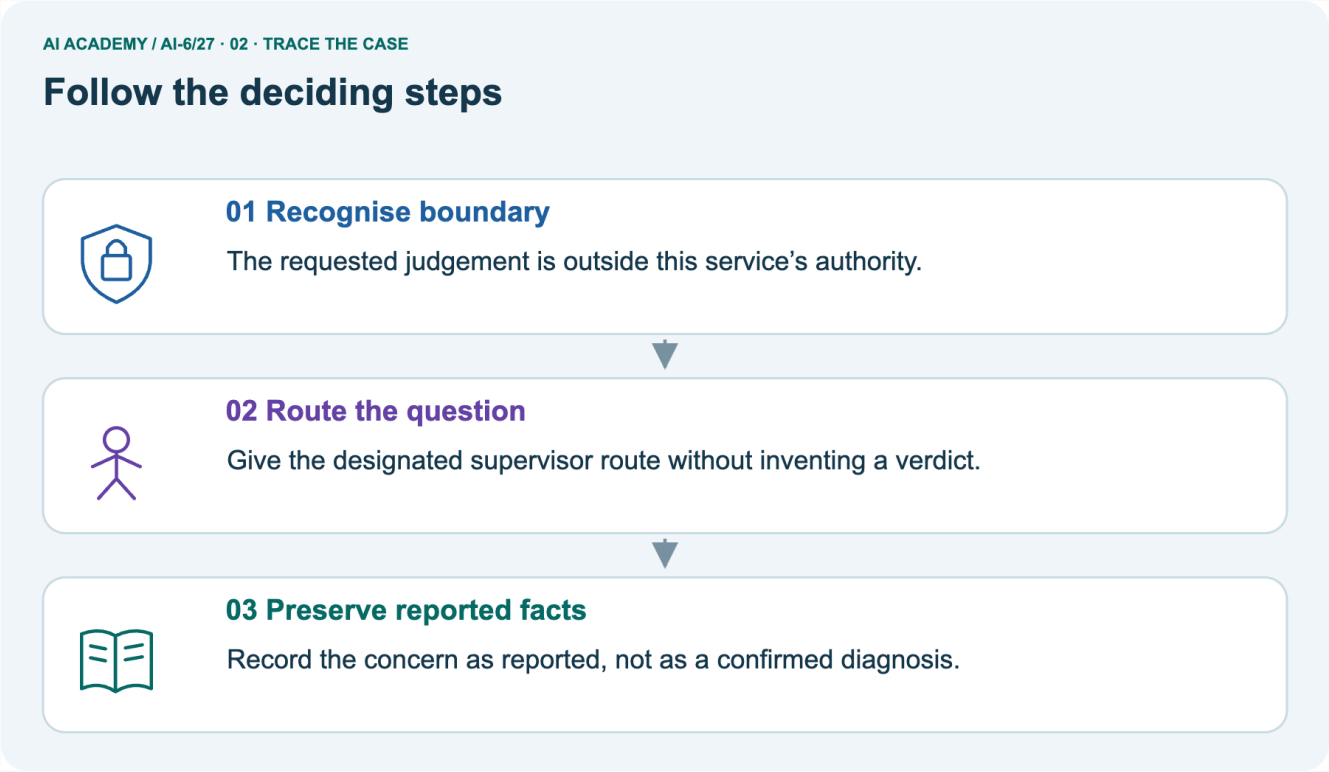
A useful human handoff. A learner asks whether a damaged laboratory bottle is safe; the bot handles lesson notes only. Supplied facts: The contract excludes equipment certification.

An assigned adult supervisor handles equipment concerns.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



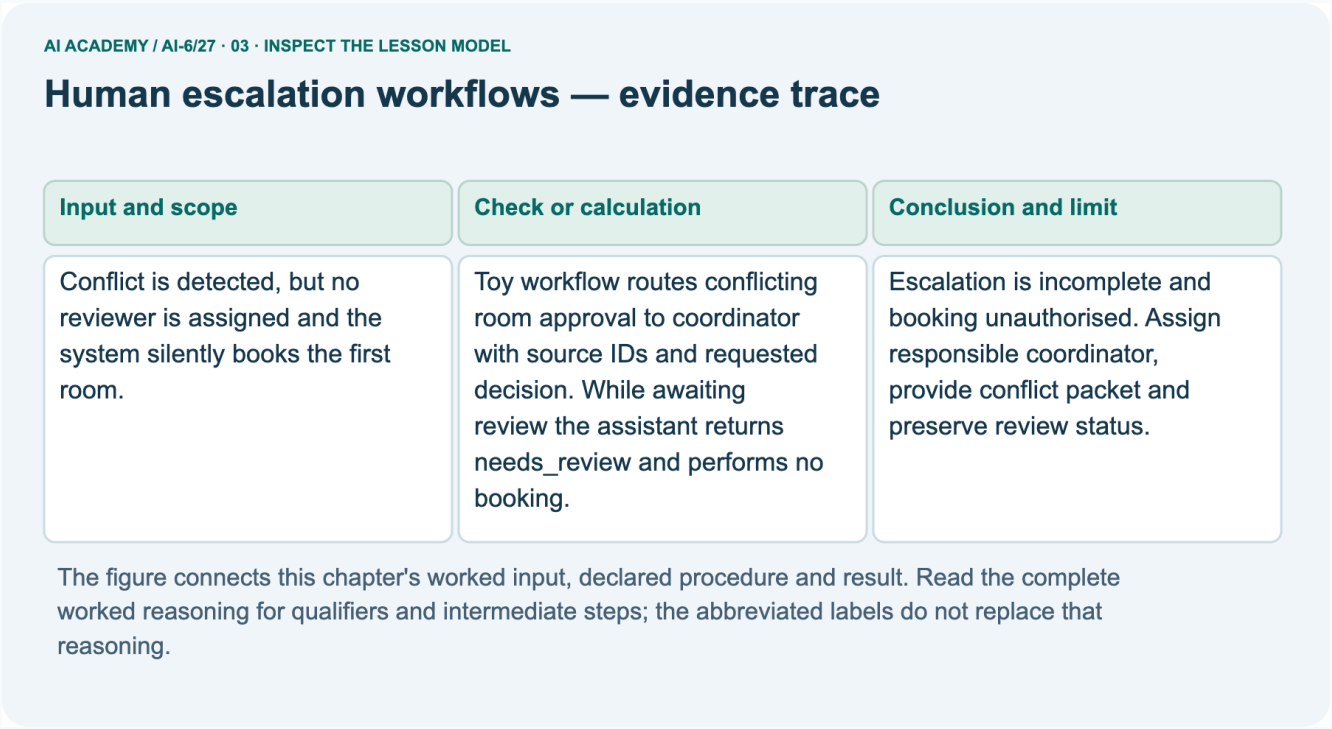
Follow the labelled picture with the matching explanation. Open the vector diagram

Recognise boundary: The requested judgement is outside this service's authority. → Route the question: Give the designated supervisor route without inventing a verdict. → Preserve reported facts: Record the concern as reported, not as a confirmed diagnosis.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Human escalation workflows — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Human escalation workflows — **evidence** trace.

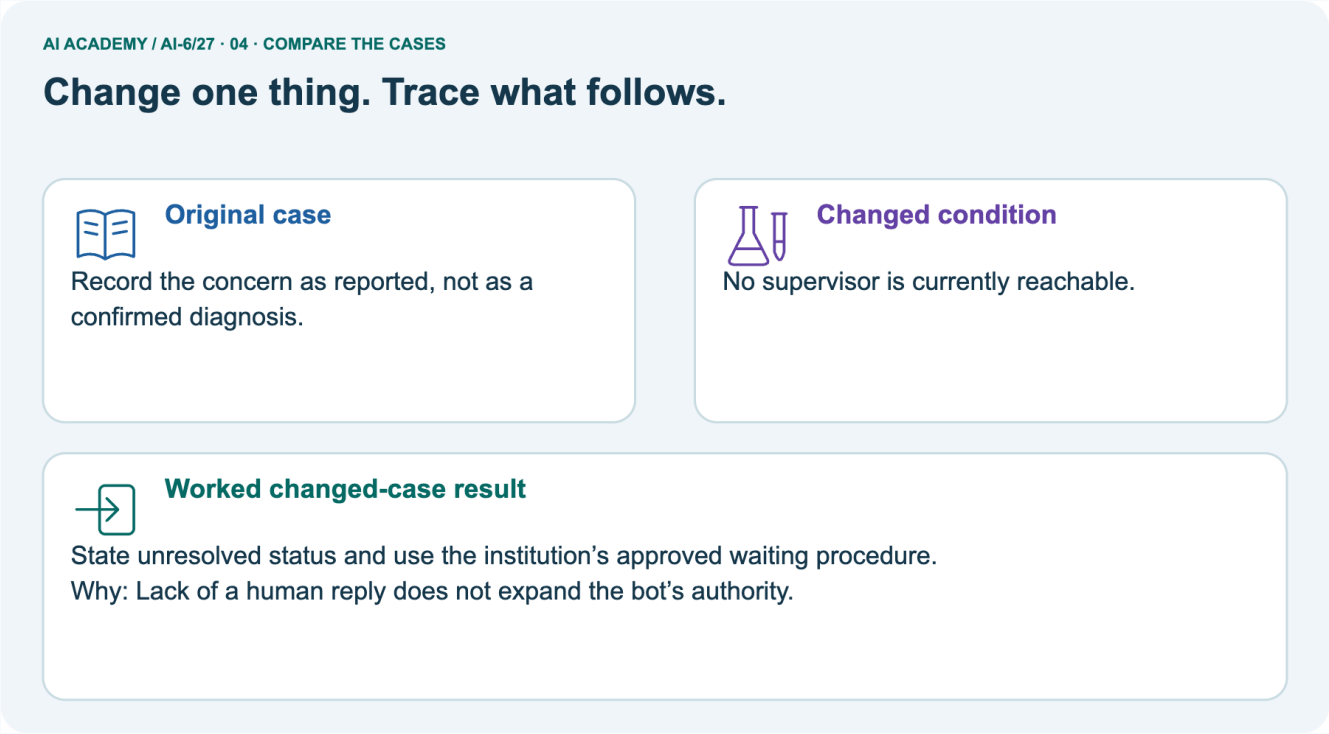
Conflict is detected, but no reviewer is assigned and the system silently books the first room. | Toy workflow routes conflicting room approval to coordinator with source IDs and requested decision.

While awaiting review the assistant returns needs_review and performs no booking. | Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

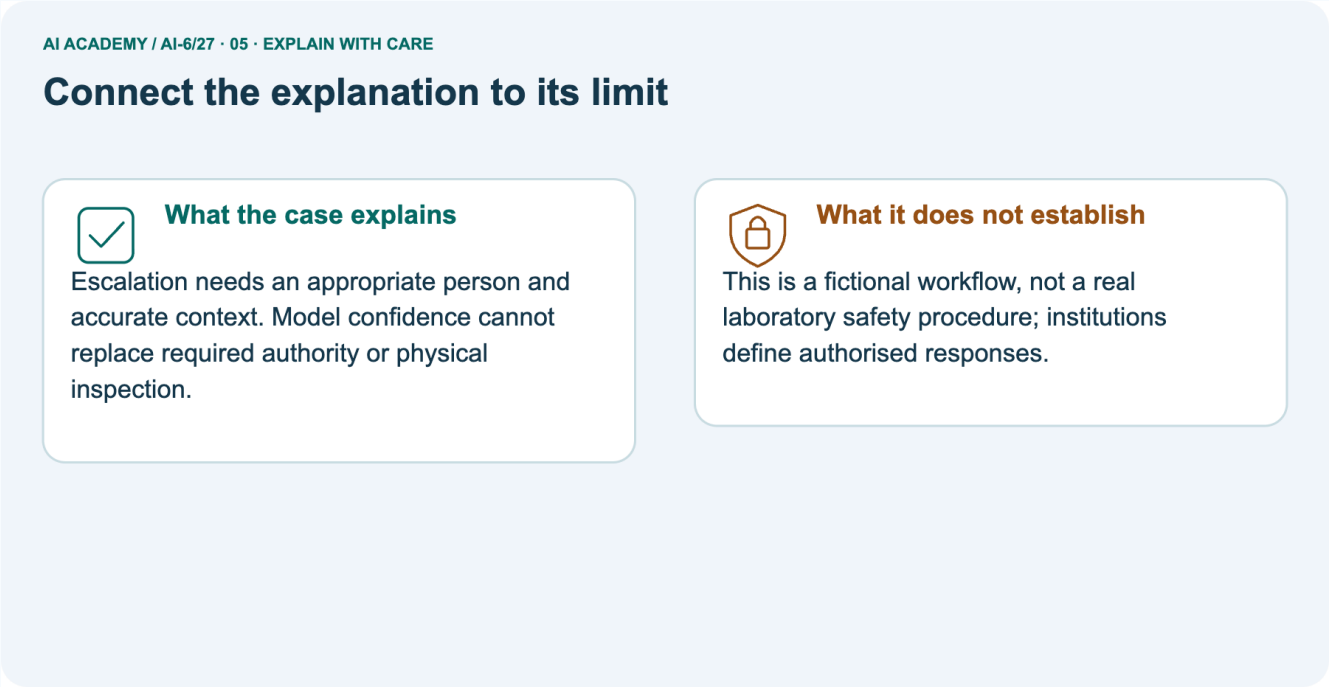
Original case: Record the concern as reported, not as a confirmed diagnosis. Changed condition: No supervisor is currently reachable.

Expected result: State unresolved status and use the institution’s approved waiting procedure. Reason: Lack of a human reply does not expand the bot’s authority.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

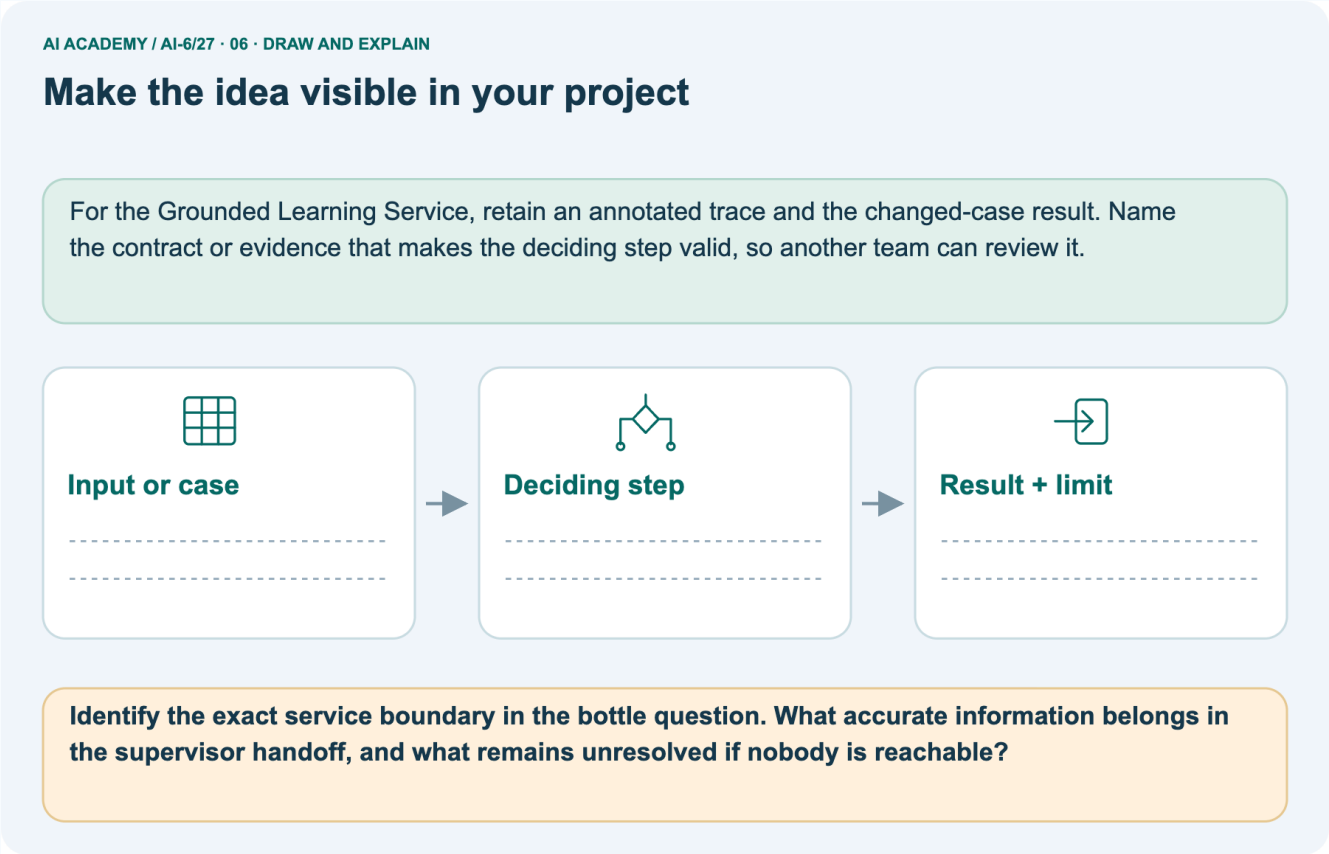
Escalation needs an appropriate person and accurate context. Model confidence cannot replace required authority or physical inspection.

Limit: This is a fictional workflow, not a real laboratory safety procedure; institutions define authorised responses.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Human escalation workflows: the case

02. Follow the deciding steps

03. Human escalation workflows — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.